Sales Domain SQL Practice Dataset & Interview-Style Questions

1. Dataset Overview

This dataset simulates a retail sales business where customers place orders for products across different regions and sales channels.

The dataset contains the following tables:

1. Customers
2. Products
3. Orders
4. Order_Items
5. Employees

These tables are designed to practice:

- GROUP BY
- CASE WHEN
- ORDER BY
- INNER / LEFT / SELF JOINS
- UNION / UNION ALL
- Window Functions
- Subqueries
- CTEs
- Aggregations
- Ranking Functions

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2. Customers Table

```
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    city VARCHAR(50),
    state VARCHAR(50),
    signup_date DATE,
    customer_segment VARCHAR(50)
);
```

```
INSERT INTO Customers VALUES
(101,'Ajeeth Kumar','Chennai','Tamil Nadu','2023-01-12','Premium'),
(102,'Rahul Sharma','Mumbai','Maharashtra','2023-02-15','Standard'),
(103,'Sneha Reddy','Hyderabad','Telangana','2023-03-10','Premium'),
```

```sql
(104,'Karan Mehta','Delhi','Delhi','2023-04-05','Standard'),
(105,'Priya Nair','Kochi','Kerala','2023-05-18','Premium'),
(106,'Vikram Singh','Bangalore','Karnataka','2023-06-21','Standard'),
(107,'Anjali Das','Kolkata','West Bengal','2023-07-01','Premium'),
(108,'Rohit Jain','Pune','Maharashtra','2023-07-19','Standard');
```

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3. Products Table

```sql
CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    price DECIMAL(10,2)
);
```

```sql
INSERT INTO Products VALUES
(201,'Laptop','Electronics',65000),
(202,'Smartphone','Electronics',30000),
(203,'Headphones','Accessories',2500),
(204,'Office Chair','Furniture',7000),
(205,'Desk Lamp','Furniture',1500),
(206,'Tablet','Electronics',25000),
(207,'Monitor','Electronics',12000),
(208,'Keyboard','Accessories',1800);
```

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4. Employees Table

```sql
CREATE TABLE Employees (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100),
    manager_id INT,
    region VARCHAR(50)
);
```

```sql
INSERT INTO Employees VALUES
(1,'Arun',NULL,'South'),
(2,'Meena',1,'South'),
(3,'Ravi',1,'West'),
(4,'Suresh',2,'North'),
(5,'Deepa',2,'East');
```

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5. Orders Table

```
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    employee_id INT,
    order_date DATE,
    sales_channel VARCHAR(50),
    order_status VARCHAR(50),
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    FOREIGN KEY (employee_id) REFERENCES Employees(employee_id)
);
```

```
INSERT INTO Orders VALUES
(1001,101,2,'2024-01-10','Online','Delivered'),
(1002,102,3,'2024-01-12','Store','Delivered'),
(1003,103,2,'2024-01-14','Online','Cancelled'),
(1004,104,4,'2024-01-16','Store','Delivered'),
(1005,105,5,'2024-01-20','Online','Delivered'),
(1006,106,3,'2024-01-22','Store','Pending'),
(1007,107,2,'2024-01-25','Online','Delivered'),
(1008,108,5,'2024-01-28','Store','Delivered'),
(1009,101,2,'2024-02-01','Online','Delivered'),
(1010,103,4,'2024-02-05','Store','Delivered');
```

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6. Order_Items Table

```
CREATE TABLE Order_Items (
    order_item_id INT PRIMARY KEY,
    order_id INT,
    product_id INT,
    quantity INT,
    discount_percent DECIMAL(5,2),
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    FOREIGN KEY (product_id) REFERENCES Products(product_id)
);
```

```
INSERT INTO Order_Items VALUES
(1,1001,201,1,10),
(2,1001,203,2,5),
(3,1002,202,1,0),
(4,1002,208,1,0),
(5,1003,206,1,15),
(6,1004,204,2,5),
```

```
(7,1005,201,1,8),
(8,1005,205,3,0),
(9,1006,207,1,10),
(10,1007,202,2,12),
(11,1008,203,4,5),
(12,1009,206,1,7),
(13,1010,204,1,0),
(14,1010,208,2,0);
```

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7. SQL Practice Questions

GROUP BY Questions

Q1.

Find total sales amount generated by each product category.

Q2.

Display the number of orders placed by each customer.

Q3.

Find the average discount percentage given for each sales channel.

Q4.

Find total quantity sold for each product.

Q5.

Show states having more than 1 premium customer.

CASE WHEN Questions

Q6.

Categorize orders into:

- High Value → amount > 50000
- Medium Value → amount between 20000 and 50000
- Low Value → amount < 20000

Q7.

Display customer names with loyalty labels:

- Premium → Gold
- Standard → Silver

Q8.

Create a column named delivery_status:

- Delivered → Success
- Pending → In Progress
- Cancelled → Failed

Q9.

Show discount category:

- 0% → No Discount
- 1–10% → Moderate Discount
- Above 10% → Heavy Discount

ORDER BY Questions

Q10.

Display top 5 highest priced products.

Q11.

Show customers ordered by signup date descending.

Q12.

Find top 3 customers who purchased highest quantity.

Q13.

Display employees based on number of handled orders.

JOIN Questions

Q14.

Display customer name, order ID, and order status.

Q15.

Display product name, quantity, and discount percentage for each order.

Q16.

Find total revenue generated by each customer.

Q17.

Display employee name along with their manager name using SELF JOIN.

Q18.

Find customers who never placed any order.

Q19.

Display all products along with total sold quantity using LEFT JOIN.

Q20.

Find revenue generated by each employee.

UNION Questions

Q21.

Combine all cities from Customers table and regions from Employees table into a single result.

Q22.

Display all Premium customers and all employees from South region using UNION.

Q23.

Use UNION ALL to display all product categories and customer segments.

WINDOW FUNCTION Questions

Q24.

Assign ROW_NUMBER to customers based on total purchase amount.

Q25.

Rank products based on total sales revenue.

Q26.

Find cumulative sales over time.

Q27.

Display previous order date for each customer using LAG().

Q28.

Find difference between current and previous purchase amount.

Q29.

Find top 2 highest revenue-generating products in each category.

Q30.

Use DENSE_RANK to rank employees by revenue.

SUBQUERY Questions

Q31.

Find customers whose total purchase amount is greater than average customer spending.

Q32.

Find products whose price is above average product price.

Q33.

Display the second highest priced product.

Q34.

Find customers who purchased Electronics products.

Q35.

Find employees who handled more orders than average employee order count.

Q36.

Display products never ordered.

INTERMEDIATE TO HARD QUESTIONS

Q37.

Find month-wise revenue and compare it with previous month revenue using window functions.

Q38.

Identify repeat customers who placed more than one order.

Q39.

Find percentage contribution of each product category to total revenue.

Q40.

Display the most sold product in each category.

Q41.

Find customers whose latest order status is Cancelled.

Q42.

Calculate running total revenue for each customer.

Q43.

Find the employee who generated maximum sales in each region.

Q44.

Find top 3 customers by revenue within each state.

Q45.

Identify orders where discount percentage is higher than average discount.

Q46.

Find products contributing to 80% of total sales.

Q47.

Display customers who bought both Electronics and Furniture products.

Q48.

Find average order value per customer.

Q49.

Find products whose sales declined compared to previous order.

Q50.

Find the longest gap between two orders for each customer.

BONUS ADVANCED QUESTIONS

Q51.

Create a CTE to calculate monthly sales trend.

Q52.

Use NTILE() to divide customers into 4 spending groups.

Q53.

Find median product price.

Q54.

Detect duplicate orders placed on same day by same customer.

Q55.

Find customers contributing more than 25% of total revenue.

Real Interview Style Scenario Questions

Q56.

Management wants to know which sales channel performs best in terms of revenue and quantity sold.

Q57.

The company plans to reduce discounts. Identify products where high discounts are not improving sales quantity.

Q58.

Find customers at risk of churn who have not placed orders in the last 30 days.

Q59.

Determine which employee consistently handles high-value orders.

Q60.

Create a leaderboard of customers based on yearly revenue.

Practice Suggestion

Recommended Order:

1. GROUP BY
2. ORDER BY
3. JOINS
4. CASE WHEN
5. SUBQUERIES
6. WINDOW FUNCTIONS
7. CTE + ADVANCED ANALYTICS

This dataset is suitable for:

- SQL Interview Preparation
- Data Analyst Practice
- Data Engineering Practice
- Business Intelligence Scenarios
- LeetCode-Style SQL Exercises
- Intermediate to Advanced SQL Learning